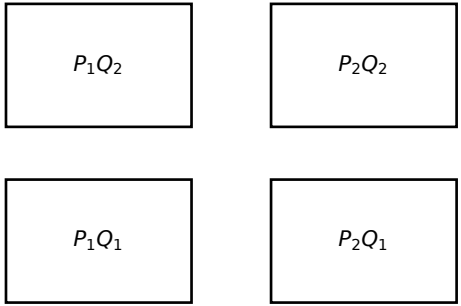
commuting resolutions

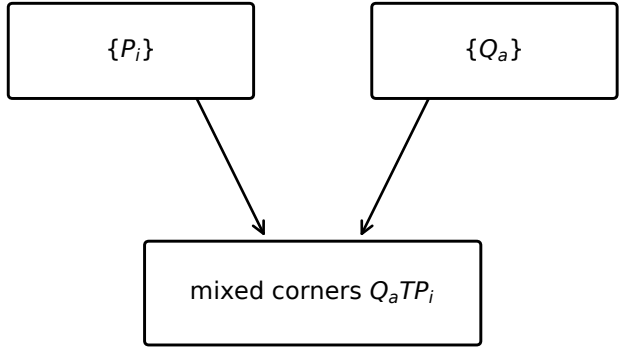
$[P_i, Q_a] = 0$



products form a common refinement

noncommuting resolutions

$[P_i, Q_a] \neq 0$



an atlas replaces a false global chart